A Positive Reciprocal-Product Criterion for the Riemann Hypothesis

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Abstract

Define

$$P(y) = \frac{\lfloor y\rfloor(\lfloor y\rfloor+1)}{y^3}\{y\}(1-\{y\}), \qquad y \geq 1,$$

and

$$T(x) = \sum_{a,b\leq\sqrt{x}} \mu(a)\mu(b)P\Big(\frac{x}{ab}\Big).$$

The Riemann hypothesis is equivalent both to

$$T(x) \ll_\varepsilon x^{1/2+\varepsilon}$$

for every $\varepsilon > 0$, and to the corresponding dyadic mean-square bound. The reduction rests on an exact truncation identity for μ already stated by Helfgott and Thompson (the case $K = 2$ of Heath-Brown's identity), applied with a weight whose periodization is the kernel P; the new content is this explicit positive kernel, its spectral structure, and the detector certificates. The kernel is entrywise nonnegative and contains no asymptotic remainder. It is nevertheless highly indefinite: along square primorials both inertias grow at least as $\log x/\log\log x$, while on a specific squarefree primorial star the centered kernel has exact inertia $(1, m-2, 0)$ and, by interlacing, the negative inertia on the full squarefree support is unbounded. We construct minimal degree-six, degree-nine, and degree-twelve nonnegative detectors which remove respectively the first one, two, and three odd collective residues. We also give a degree-fifteen detector, minimal in the same endpoint-flat polynomial class, which removes the first four odd collective residues; its positivity is certified by an exact shifted Faulhaber–Bernstein expansion. An analogous degree-eighteen certificate removes the first five odd residues. These are equivalent criteria and structural results, not a proof of RH.

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1. Introduction

Square-root bounds for Möbius sums are classical reformulations of RH. Here one such bound is converted into a finite reciprocal-product quadratic form whose entries are nonnegative. With $\lfloor y\rfloor$ and $\{y\}$ the integer and fractional parts, put

$$P(y) = \frac{\lfloor y\rfloor(\lfloor y\rfloor+1)}{y^3}\,\{y\}(1-\{y\}) \qquad (y \geq 1), \qquad T(x) = \sum_{a,b\leq\sqrt{x}} \mu(a)\mu(b)\,P\Big(\frac{x}{ab}\Big) \qquad (x \geq 1).$$

The main theorem (Section 4) states that RH is equivalent to $T(x) \ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon > 0$, and to the corresponding dyadic mean-square bound. The positivity is literal but is not positive semidefiniteness; the construction therefore isolates, rather than solves, the required signed cancellation.

Positive Farey quadratic forms give other RH criteria; see Kanemitsu and Yoshimoto (1996) [1]. Their Theorem 3, called Franel's formula in the original paper, is a nonnegative Farey square identity whose arithmetic form is a double sum in $M(x/m)M(x/n)(m,n)^2/(mn)$. It therefore provides the closest positive-quadratic precedent, but not the $a, b \leq \sqrt{x}$, $x/(ab)$ kernel below. More precisely, in the notation of Báez–Duarte's Theorem 3.1 [2] (operator G_ϕ as in

his (3.1); equivalence (3.14)), take $F(t) = t^{-1}V(1/t)$ and $\phi(t) = tF'(t)$. Then direct calculation gives $G_\phi(x) = U_V(x)/x$ (and $\widehat{\phi}(s) = sK_V(s+1)$, with the corresponding Mellin convention). Thus the pointwise detector equivalence here is a special case of that general Möbius-convolution theorem and makes no new contribution. Liflandsky's [3] (Theorem 1.3(a)) unconditional implication from the stated no-ε discrete-Laplace bound to RH has a converse (Theorem 1.3(b) of [3]) only under RH together with his Hypotheses (S) and (H). Verjovsky's [4] Laplace L^p criterion (Theorem 3.1) uses Mellin continuation via Hölder's inequality. His local Fourier-moment criterion (Theorem 4.1 of [4]) uses derivative bounds for Fourier polynomials and the persistence of peaks (Lemma 9.1 and Proposition 9.2 of [4]), not a general L^2-to-pointwise principle. None of these analytic mechanisms is claimed here as new. Statistical aspects of the Möbius sequence were studied by Good and Churchhouse (1968) [5].

The exact truncation identity (4.1) on which Section 4 rests is not new: it is stated explicitly by Helfgott and Thompson [7, Section 2, equation (2.1)], with cutoff $u \geq \sqrt{x}$, who identify it as the case $K = 2$ of Heath-Brown's identity for μ and note that it also follows from Vaughan's identity. The weighted identity (4.2) is its application with the weight V. A full-text comparison against the closest criterion, Farey, and Möbius-convolution precedents did not locate the positive kernel (3.2), its bounded transfer (4.3)–(4.4), or the five certified residue-removing detectors of Section 8. Accordingly the novelty claim of this paper is restricted to the explicit positive kernel, its inertia, the five residue-removing detectors and their certificates; their minimality (Proposition 8.1) uses the uniform argument of [8, Theorem 2.1], published after the first version of this paper. The exact truncation mechanism (4.1), the detector theorem itself, the equivalence $M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \iff$ RH, and the general use of Mellin transforms are background. This is a bounded priority statement, not an assertion that all RH criteria have been exhaustively classified.

We also compared the claim with rStation's recent Mellin-filter work [6]. That paper proves quantitative noncancellation and perturbation bounds for smoothed Nyman–Beurling filters. The present contribution concerns an exact finite reciprocal-product identity, its inertia, and polynomial periodization certificates; it does not claim novelty for the general Mellin noncancellation mechanism.

2. A zero-free Mellin detector

Let $W \in BV(0,\infty)$ be supported in $(0, B]$ after zero extension. Set

$$U_W(x) = \sum_{n\geq 1} \mu(n) W(n/x), \qquad K_W(s) = \int_0^\infty W(t) t^{s-1}\, dt.$$

For $\Re s > 1$, absolute convergence and $t = n/x$ give

$$\int_0^\infty U_W(x) x^{-s-1}\, dx = \frac{K_W(s)}{\zeta(s)}. \tag{2.1}$$

Detector theorem. If

$$K_W(s) \neq 0 \qquad (1/2 < \Re s < 1), \tag{2.2}$$

then RH is equivalent to

$$\forall \varepsilon > 0, \qquad \int_Y^{2Y} |U_W(x)|^2\, dx \ll_{W,\varepsilon} Y^{2+\varepsilon} \qquad (Y \geq 1). \tag{2.3}$$

Proof. RH gives

$$M(X) = \sum_{n\leq X} \mu(n) = O_\delta(X^{1/2+\delta})$$

for every $\delta > 0$. Abel summation, including all jumps in the total variation of W, yields

$$|U_W(x)| \ll_W \sup_{n\leq Bx} |M(n)| \ll_{W,\delta} x^{1/2+\delta}.$$

Taking $\delta = \varepsilon/2$ proves (2.3).

Conversely, fix $\sigma_0 > 1/2$ and choose $0 < \varepsilon < 2\sigma_0 - 1$. Dyadic summation gives

$$\int_{1/B}^\infty |U_W(x)|^2 x^{-2\sigma_0-1}\, dx < \infty.$$

If $\Re s > \sigma_0$, Cauchy–Schwarz with the additional factor $x^{-(\Re s-\sigma_0)}$ makes the left side of (2.1) absolutely convergent, locally uniformly in s. Since $\sigma_0 > 1/2$ is arbitrary, it is holomorphic on $\Re s > 1/2$. With $Z(s) = (s-1)\zeta(s)$, continuation of

$$Z(s)\int_0^\infty U_W(x)x^{-s-1}\,dx = (s-1)K_W(s)$$

shows that a zero of ζ in $1/2 < \Re s < 1$ would force a zero of K_W there, with at least the same multiplicity. This contradicts (2.2). The classical zero-free half-plane $\Re s \geq 1$, the functional equation, and the symmetry $\rho \mapsto 1-\overline{\rho}$ put every nontrivial zero on the critical line. ∎

3. The endpoint-flat kernel

Define

$$V(u) = \begin{cases} -2u + 6u^2 - 4u^3, & 0 < u < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Its zero extension is continuous and of bounded variation, $V(0+) = V(1-) = 0$, and $\int_0^1 V = 0$. Moreover,

$$K_V(s) = -\frac{2}{s+1} + \frac{6}{s+2} - \frac{4}{s+3} = \frac{2(s-1)}{(s+1)(s+2)(s+3)}, \tag{3.1}$$

which is zero-free in $1/2 < \Re s < 1$.

Faulhaber factorization. With P as defined in Section 1, for $y \geq 1$,

$$\sum_{c\leq y} V(c/y) = P(y) \geq 0. \tag{3.2}$$

Indeed, writing $N = \lfloor y \rfloor$ and $r = \{y\}$, the first three power-sum formulas give

$$\sum_{c\leq y} V(c/y) = -\frac{N(N+1)}{y} + \frac{N(N+1)(2N+1)}{y^2} - \frac{N^2(N+1)^2}{y^3}.$$

After factoring $N(N+1)/y^3$, the remaining polynomial is

$$-y^2 + (2N+1)y - N(N+1) = r(1-r).$$

4. Exact truncation and the RH criterion

For real $x \geq 1$, put

$$\nu_x(n) = \mu(n)\mathbf{1}_{n\leq\sqrt{x}}.$$

For every integer $n \leq x$,

$$\mu(n) = 2\nu_x(n) - (\nu_x * \nu_x * 1)(n). \tag{4.1}$$

This is the identity (2.1) of Helfgott and Thompson [7] with $u = \sqrt{x}$, the case $K = 2$ of Heath-Brown's identity; we include the short proof for completeness.

To prove this, set $z = \sqrt{x}$ and $r = \delta - \nu_x * 1$. Then $r(m) = 0$ for every integer $m \leq z$. If $n \leq z^2$, every factorization $n = ab$ has $a \leq z$ or $b \leq z$; hence $(r * r)(n) = 0$. Since $\mu * 1 = \delta$,

$$0 = (\mu * r * r)(n) = \mu(n) - 2\nu_x(n) + (\nu_x * \nu_x * 1)(n).$$

Let

$$U(x) = \sum_n \mu(n)V(n/x).$$

Recall $T(x) = \sum_{a,b\leq\sqrt{x}} \mu(a)\mu(b)P(x/(ab))$ from Section 1. Multiplying (4.1) by $V(n/x)$, summing, and using (3.2) gives

$$U(x) = 2A(x) - T(x), \qquad A(x) = \sum_{n\leq\sqrt{x}} \mu(n)V(n/x). \tag{4.2}$$

There is no endpoint term because $V(1) = 0$. If $N = \lfloor\sqrt{x}\rfloor$, then

$$|A(x)| \le \frac{2}{x}\sum_{n\le N} n + \frac{6}{x^2}\sum_{n\le N} n^2 + \frac{4}{x^3}\sum_{n\le N} n^3 \le 12. \tag{4.3}$$

Consequently,

$$\boxed{|T(x) + U(x)| \le 24 \qquad (x \ge 1).} \tag{4.4}$$

Main theorem. The following are equivalent:

1. RH;
2. for every $\varepsilon > 0$,
$$T(x) \ll_\varepsilon x^{1/2+\varepsilon} \qquad (x \ge 1);$$
3. for every $\varepsilon > 0$,
$$\int_Y^{2Y} |T(x)|^2\,dx \ll_\varepsilon Y^{2+\varepsilon} \qquad (Y \ge 1).$$

The detector theorem gives the mean-square equivalence for U. More precisely, (4.4) and $|a+b|^2 \le 2|a|^2 + 2|b|^2$ give

$$\int_Y^{2Y} |T(x)|^2\,dx \le 2\int_Y^{2Y} |U(x)|^2\,dx + O(Y)$$

and, in the reverse direction,

$$\int_Y^{2Y} |U(x)|^2\,dx \le 2\int_Y^{2Y} |T(x)|^2\,dx + O(Y).$$

Thus the bounded error transfers the required estimate in both directions. RH also gives the pointwise bound for U by Abel summation. Conversely, for a desired dyadic exponent $\eta > 0$, apply the pointwise bound with $\varepsilon = \eta/2$, square and integrate.

The quantifier "for every $\varepsilon > 0$" is essential. The pointwise and mean-square versions are separately equivalent through RH; no general L^2-to-pointwise implication is asserted.

Corollary 4.1 (critical endpoint). Unconditionally,

$$T(x) \ne o(x^{1/2}), \qquad \int_1^\infty \frac{|T(x)|^2}{x^2}\,dx = \infty.$$

Moreover, if $T(x) = O(x^{1/2})$, then RH holds and every nontrivial zero of ζ is simple.

This is an application of classical Mellin arguments, parallel to Proposition 4.1, Theorem 4.2 (equation (4.3)) and Theorem 4.1 of [2] (stated there for mellin-proper ϕ); it is not claimed as new, and the self-contained proof below does not use [2]. For the proof, note that $U(x) = 0$ for $x \le 1$ and put $H(s) = \int_1^\infty U(x)x^{-s-1}\,dx$. Under each of the three hypotheses below (transferred from T to U by (4.4), bounded functions lying in $L^2([1,\infty), x^{-2}dx)$), the integral converges absolutely and locally uniformly on $\Re s > 1/2$, so by (2.1) $H = K_V/\zeta$ there. Let $\rho = \frac{1}{2} + i\gamma$ be a zero of ζ of multiplicity $k \ge 1$ (such zeros exist), and let $\delta \downarrow 0$. By (3.1), $K_V(\rho) \ne 0$, so $|H(\rho+\delta)| \asymp \delta^{-k}$.

- If $U(x) = o(x^{1/2})$, split the integral at a large X: $\delta|H(\rho+\delta)| \le \delta\, O_X(1) + \sup_{x\ge X} |U(x)|x^{-1/2}$, so $\delta H(\rho+\delta) \to 0$, contradicting $k \ge 1$.
- If $\int_1^\infty |U(x)|^2 x^{-2}\,dx < \infty$, Cauchy–Schwarz gives $|H(\rho+\delta)| \le \left(\int_1^\infty |U|^2 x^{-2}dx\right)^{1/2}(2\delta)^{-1/2}$, again contradicting $k \ge 1$.
- If $U(x) = O(x^{1/2})$, the main theorem gives RH, and $|H(\rho+\delta)| = O(\delta^{-1})$ forces $k = 1$.

Divergence of the weighted integral does not exclude the dyadic bound $\int_Y^{2Y} |T(x)|^2\,dx \ll Y^2$; the two statements are not interchangeable.

The proof-bearing identity is (4.1). Its quantifiers are: for every real $x \ge 1$ and every integer $n \le x$, with the cutoff in ν_x depending on that same x. After multiplication by $V(n/x)$, the support condition and $V(1) = 0$ justify the finite rearrangement into (4.2); no limiting interchange or asymptotic Euler summation occurs. Thus the theorem is an equivalence of an unproved family of estimates with RH, not an estimate for T and not a proof of RH.

5. Spectral geometry

For $1 \le a, b \le \sqrt{x}$, write

$$K_x(a,b) = P(x/(ab)), \qquad C_x(a,b) = K_x(a,b) - \frac{ab}{6x}.$$

The entrywise nonnegative matrix K_x is not generally positive semidefinite. For every $k \ge 1$, the indices $(3k, 2k)$ at $x = 9k^2$ give the principal block

$$\begin{pmatrix} 0 & 4/27 \\ 4/27 & 8/81 \end{pmatrix}, \qquad \det = -\frac{16}{729}. \tag{5.1}$$

The indefiniteness has growing rank. Let $p_1 < \cdots < p_m$ be the first m primes, $Q = p_m^{\#}$, and $x = Q^2$. For $m \ge 3$, on the indices

$$a_k = p_k, \qquad b_k = p_k^2,$$

all cross-block products divide Q^2, so their kernel entries vanish. The resulting principal submatrix is

$$\bigoplus_{k=1}^{m} \begin{pmatrix} 0 & c_k \\ c_k & d_k \end{pmatrix}, \qquad c_k > 0,\ d_k \ge 0.$$

Every block has one eigenvalue of each sign. Since C_x is obtained there by subtracting a rank-one positive semidefinite matrix,

$$n_{\pm}(K_x) \ge m, \qquad n_-(C_x) \ge m, \qquad n_+(C_x) \ge m - 1. \tag{5.2}$$

As $\log x = (2 + o(1))m \log m$, both inertias grow at least as $\log x / \log\log x$ along this subsequence.

6. Exact squarefree centered inertia

Let $x = p_m^{\#}$ and take the squarefree indices $a_j = 2p_j$, $2 \le j \le m$. For $m \ge 6$, the corresponding principal matrices satisfy

$$\boxed{(n_+, n_-, n_0)(K) = (n_+, n_-, n_0)(C) = (1, m-2, 0).} \tag{6.1}$$

Indeed, with $D_{jj} = p_j/\sqrt{x}$, the half-integer formula for P and the diagonal calculation give

$$K = D(J - E)D, \qquad C = D\left(\frac{1}{3}J - E\right)D, \tag{6.2}$$

where $J = \mathbf{1}\mathbf{1}^T$ and

$$E_{ij} = \frac{4p_i^2 p_j^2}{x^2} \quad (i \ne j), \qquad E_{jj} = 1 - 4y_j P(y_j), \qquad y_j = \frac{x}{4p_j^2}. \tag{6.3}$$

Since $P \ge 0$, these entries satisfy

$$1 \ge E_{jj} > p_m^{-2}, \qquad \sum_{k \ne j} E_{jk} < p_m^{-2}, \tag{6.4}$$

so $E \succ 0$ by strict diagonal dominance. To verify the diagonal bound, write $y_j = N_j + r_j$. Since $x/(2p_j)$ is odd and not divisible by p_j, one has $d_j := 1 - 2r_j = k_j/p_j$, where k_j is a nonzero even integer with $2 \le |k_j| \le p_j - 1$. Direct substitution in P gives

$$E_{jj} = d_j^2 - \frac{(1-d_j^2)d_j}{y_j} + \frac{(1-d_j^2)^2}{4y_j^2} \ge \frac{k_j^2}{p_j^2} - \frac{4|k_j|p_j}{x}.$$

For $|k_j| = 2$, this is exactly $4/p_j^2 - 8p_j/x$. If $|k_j| > 2$, the same lower bound follows from

$$\frac{k_j^2 - 4}{p_j^2} \ge \frac{4p_j(|k_j| - 2)}{x},$$

which is equivalent to $x(|k_j| + 2) \ge 4p_j^3$. This holds because $|k_j| + 2 \ge 6$, $p_j \le p_m$, and $x > (8/3)p_m^3$. Consequently

$$E_{jj} \ge \frac{4}{p_j^2} - \frac{8p_j}{x} > p_m^{-2}.$$

For the off-diagonal sum,

$$\sum_{k\neq j} E_{jk} \le \frac{4(m-2)p_m^4}{x^2} < p_m^{-2}.$$

The last inequalities reduce to $x > (8/3)p_m^3$ and $x^2 > 4(m-2)p_m^6$. They hold at $m = 6$. For the induction step, put $p = p_m$, $p' = p_{m+1}$, and use Bertrand's bound $p' < 2p$: $p^3 > p'^2$ and $(m-2)p^2 > 16(m-1)$ for $m \ge 6$. Multiplication by p' preserves both inequalities. Put $n = m-1$ and $q = \mathbf{1}^T E^{-1}\mathbf{1}$. Congruence by $E^{-1/2}$ turns the middle matrix for C into

$$\frac{1}{3}uu^T - I, \qquad u = E^{-1/2}\mathbf{1}.$$

It has $n-1$ negative eigenvalues, while the remaining sign is that of $q/3-1$. Cauchy–Schwarz and (6.4) give

$$q \ge \frac{n^2}{\mathbf{1}^T E\mathbf{1}} > \frac{n}{1+p_m^{-2}} \ge \frac{5}{1+13^{-2}} > 3.$$

This proves (6.1) for C; replacing $1/3$ by 1 proves it for K.

Corollary 6.1 (squarefree support). Let $I_x = \{a \le \sqrt{x} : \mu(a) \neq 0\}$, and let K_x^{sf} and C_x^{sf} be the principal submatrices of K_x and C_x on I_x. For $x = p_m^{\#}$ and $m \ge 6$,

$$n_-(K_x^{\mathrm{sf}}) \ge m-2, \qquad n_-(C_x^{\mathrm{sf}}) \ge m-2, \qquad n_+(K_x^{\mathrm{sf}}) \ge 1, \qquad n_+(C_x^{\mathrm{sf}}) \ge 1.$$

Indeed, each $2p_j$ ($2 \le j \le m$) is squarefree and satisfies $2p_j \le 2p_m < \sqrt{x}$ because $x > (8/3)p_m^3 > 4p_m^2$; hence the matrices of (6.1) are principal submatrices of K_x^{sf} and C_x^{sf}, and Cauchy interlacing gives the bounds. Since $\log x = \vartheta(p_m) = (1+o(1))m\log m$, the negative inertia on the full squarefree support is at least $(1+o(1))\log x/\log\log x$ along primorials. This concerns the inertia of the form, not its value at the Möbius vector.

7. Low-degree positivity cone

For

$$V(u) = \sum_{k=1}^{d} v_k u^k, \qquad V(0) = V(1) = \int_0^1 V(u)\,du = 0,$$

put $P_V(y) = \sum_{c\le y} V(c/y)$. For $d = 3$,

$$P_V(y) \ge 0 \quad (y \ge 1) \iff v_3 \le 0. \tag{7.1}$$

For $d = 4$, the endpoint and mean constraints give

$$v_1 = \frac{v_3}{2} + \frac{4v_4}{5}, \qquad v_2 = -\frac{3v_3}{2} - \frac{9v_4}{5},$$

and the exact classification is

$$\boxed{P_V(y) \ge 0 \quad (y \ge 1) \iff v_4 \ge 0, \qquad v_3 \le -\frac{12}{5}v_4.} \tag{7.2}$$

Necessity follows from

$$P_V(N) = \frac{v_4(N^2-1)}{30N^3}$$

and the $N = 1$, $r \downarrow 0$ boundary. For sufficiency write $v_4 = w \ge 0$, $v_3 = -12w/5 - t$, $t \ge 0$. With $y = N + r$, the numerator of $60P_V(N+r)(N+r)^4/[N(N+1)]$ becomes

$$15tr(N+r)(1-r) + w\big(2(N-1) + 6r^2(3-2r)\big) \ge 0.$$

Thus endpoint flatness alone does not imply positivity. For example,

$$V(u) = -\frac{4}{5}u + \frac{9}{5}u^2 - u^4$$

has zero endpoints and zero mean, but

$$P_V(N) = -\frac{N^2-1}{30N^3} < 0 \qquad (N \ge 2).$$

8. A hierarchy of positive residue-removing detectors

The cubic kernel contains the collective term

$$\frac{1}{6x}\left(\sum_{a\le\sqrt{x}} a\mu(a)\right)^2,$$

coming from the $s=-1$ Mellin residue. This term is not forced by entrywise kernel positivity.

We first record the endpoint lemma used for the minimality statements. Fix $\ell \ge 1$. Suppose that V is a real polynomial with zero endpoints and zero mean, that

$$[u]V = [u^3]V = \cdots = [u^{2\ell-1}]V = 0, \qquad P_V(y) \ge 0 \quad (y \ge 1).$$

For $y = N + r$, shifted Euler–Faulhaber summation gives

$$P_V(N+r) = \sum_{k\ge1} \frac{B_k(1-r)V^{(k-1)}(1) - B_k V^{(k-1)}(0)}{k!(N+r)^{k-1}}. \tag{8.0}$$

For $2 \le k \le 2\ell+1$, the lower-endpoint term vanishes: when k is even this follows from the displayed odd-coefficient conditions, and when $k > 1$ is odd it follows from $B_k = 0$. Each Bernoulli polynomial B_k with $k \ge 2$ is nonzero and has $\int_0^1 B_k = 0$, so $B_k(1-r)$ changes sign on $0 < r < 1$. Therefore the first nonzero upper-endpoint derivative would make (8.0) negative for a fixed r and all sufficiently large N. Hence

$$u^2(1-u)^{2\ell+1} \mid V. \tag{8.0a}$$

Proposition 8.1 (minimal degree for $\ell \le 5$; [8, Theorem 2.1]). Let $1 \le \ell \le 5$ and let $V \ne 0$ be a real polynomial with $V(0) = V(1) = \int_0^1 V = 0$, $[u^{2j-1}]V = 0$ for $1 \le j \le \ell$, and $P_V(y) \ge 0$ for $y \ge 1$. Then $\deg V \ge 3\ell + 3$.

This bound is proved for every $\ell \ge 1$ in [8, Theorem 2.1], which credits the five finite constructions of the first version of this paper; the denominator in (8.0b) below is that of [8, (1.2)–(1.3)]. The five finite minimality results were first obtained here by separate arguments; the uniform proof below is the argument of [8], reproduced for completeness.

Proof. Let $d = \deg V$. By (8.0a), $d \ge 2\ell + 3$. Write $V = \sum_{j=1}^{d} v_j u^j$, so $K_V(s) = \sum_j v_j/(s+j)$ with $v_j = 0$ for odd $j < 2\ell$. With $D(s) = \prod(s+j)$ over $1 \le j \le d$, $j \notin \{1, 3, \dots, 2\ell-1\}$, we get $K_V = A/D$ with $\deg D = d - \ell$ and $A \ne 0$ a polynomial (partial fractions are unique). Writing $V = (1-u)^{2\ell+1}H$ by (8.0a), for real $s > 0$

$$|K_V(s)| \le \|H\|_\infty \int_0^1 u^{s-1}(1-u)^{2\ell+1}\,du = \|H\|_\infty \frac{(2\ell+1)!}{s(s+1)\cdots(s+2\ell+1)} = O(s^{-2\ell-2}),$$

hence $\deg A \le d - 3\ell - 2$. Zero mean gives $A(1) = D(1)K_V(1) = 0$, so $\deg A \ge 1$ and $d \ge 3\ell + 3$. ∎

For each of the five detectors constructed below, with $d = 3\ell + 3$, the Mellin transform has the form

$$K_V(s) = \frac{C(s-1)}{\prod_{j=1}^{\ell}(s+2j)\prod_{k=2\ell+1}^{3\ell+3}(s+k)}, \qquad C \ne 0. \tag{8.0b}$$

The endpoint lemma (8.0a) holds for every $\ell \ge 1$, and so does the degree bound [8, Theorem 2.1]; Proposition 8.1 is stated only for the five levels at which detectors are constructed and certified in this paper.

Now let

$$V_6(u) = u^2(1-u)^3(7u-3) = -3u^2 + 16u^3 - 30u^4 + 24u^5 - 7u^6. \tag{8.1}$$

Then

$$K_{V_6}(s) = \frac{24(s-1)}{(s+2)(s+3)(s+4)(s+5)(s+6)}, \tag{8.2}$$

so the detector remains zero-free in $1/2 < \Re s < 1$, while $\operatorname{Res}_{s=-1} K_{V_6} = 0$.

If $y = N + r$, $N = \lfloor y \rfloor \geq 1$, $0 \leq r < 1$, its exact Faulhaber kernel is

$$P_{V_6}(N + r) = \frac{N(N+1)}{6(N+r)^6} \left\{ (N-1)[1 - 6r^2(1-r)^2] + 3r^3(4 - 3r) \right\}. \tag{8.3}$$

Both terms in braces are nonnegative because $r^2(1-r)^2 \leq 1/16$. Hence $P_{V_6}(y) \geq 0$ on the entire half-line. The same exact truncation proof gives another positive reciprocal-product criterion for RH, now with transfer error $O(x^{-1/2})$, because $V_6(u) = O(u^2)$ at zero.

To state the residue decompositions, write

$$S_j(z) = \sum_{n \leq z} n^j \mu(n).$$

If $V(u) = \sum_{j=1}^{d} v_j u^j$, (8.0) gives the exact collective decomposition

$$P_V(y) = \sum_{j=1}^{d} v_j \zeta(-j) y^{-j} + R_V(y), \qquad R_V(y) = \sum_{j=1}^{d} \frac{V^{(j)}(1)}{(j+1)!} B_{j+1}(1 - \{y\}) y^{-j}.$$

Accordingly, for the reciprocal-product form $T_V(x) = \sum_{a,b \leq \sqrt{x}} \mu(a)\mu(b) P_V(x/(ab))$, define

$$T_V^{(0)}(x) = \sum_{a,b \leq \sqrt{x}} \mu(a)\mu(b) R_V\Big(\frac{x}{ab}\Big),$$

so that

$$T_V(x) = \sum_{j=1}^{d} v_j \zeta(-j) x^{-j} S_j(\sqrt{x})^2 + T_V^{(0)}(x).$$

The Bernoulli factors in R_V have zero mean as functions of the fractional part. This periodic zero-mean fact does not provide any arithmetic mean-square control of $T_V^{(0)}$.

The remaining nonzero polynomial residues are

$$T_{V_6}(x) = \frac{2S_3(\sqrt{x})^2}{15x^3} - \frac{2S_5(\sqrt{x})^2}{21x^5} + T_{V_6}^{(0)}(x). \tag{8.4}$$

Thus the collective skeleton moves from (S_1, S_3) to (S_3, S_5). Degree six is minimal by Proposition 8.1 with $\ell = 1$; (8.2) is the case $\ell = 1$ of (8.0b).

The next level is also explicit and sharp. Define

$$V_9(u) = u^2(1-u)^5(20u^2 - 5u - 1). \tag{8.5}$$

Then

$$K_{V_9}(s) = \frac{1680(s-1)}{(s+2)(s+4)(s+5)(s+6)(s+7)(s+8)(s+9)}. \tag{8.6}$$

Thus both $s = -1$ and $s = -3$ residues disappear, while the critical strip remains zero-free. If $y = N + r$, $m = N - 1$, the exact kernel has the form

$$P_{V_9}(N + r) = \frac{N(N+1)}{6(N+r)^9} Q_N(r), \tag{8.7}$$

where, in the Bernstein basis $\beta_{k,7}(r) = \binom{7}{k} r^k (1-r)^{7-k}$,

$$Q_N(r) = \sum_{k=0}^{7} \Big(c_k^{(0)} + m c_k^{(1)} + m^2 c_k^{(2)} \Big) \beta_{k,7}(r), \tag{8.8}$$

with

$$\begin{aligned} c^{(0)} &= (0,0,0,0,0,2,9,18),\\ c^{(1)} &= (3,6,28/3,12,12,32/3,12,15),\\ c^{(2)} &= (3,3,10/3,4,4,10/3,3,3). \end{aligned} \tag{8.9}$$

Every coefficient is nonnegative, proving $P_{V_9}(y) \geq 0$ on the whole half-line. Its collective skeleton begins two odd residues later:

$$T_{V_9}(x) = \frac{5S_5(\sqrt{x})^2}{9x^5} - \frac{14S_7(\sqrt{x})^2}{15x^7} + \frac{5S_9(\sqrt{x})^2}{33x^9} + T^{(0)}_{V_9}(x). \tag{8.10}$$

Degree nine is minimal by Proposition 8.1 with $\ell = 2$.

The third level continues the same sharp degree progression. Let

$$V_{12}(u) = u^2(1-u)^7(91u^3 - 3u^2 - 7u - 1). \tag{8.11}$$

Then

$$K_{V_{12}}(s) = \frac{403200(s-1)}{(s+2)(s+4)(s+6)(s+7)(s+8)(s+9)(s+10)(s+11)(s+12)}. \tag{8.12}$$

An exact degree-ten Bernstein expansion, with coefficients polynomial in $m = N-1$ in the nonnegative falling-factorial basis

$$1, \qquad m, \qquad m(m-1), \qquad m(m-1)(m-2),$$

has only nonnegative rational coefficients; Supplementary Table S1 records the complete array. Hence $P_{V_{12}}(y) \geq 0$ for every $y \geq 1$. Its first three odd residues vanish, and

$$T_{V_{12}}(x) = \frac{112S_7(\sqrt{x})^2}{15x^7} - \frac{800S_9(\sqrt{x})^2}{33x^9} + \frac{11056S_{11}(\sqrt{x})^2}{819x^{11}} + T^{(0)}_{V_{12}}(x). \tag{8.13}$$

By Proposition 8.1 with $\ell = 3$, degree twelve is minimal.

The next four-residue construction is already available in degree fifteen. Let

$$V_{15}(u) = u^2(1-u)^9\big(1024u^4 + 207u^3 - 57u^2 - 27u - 3\big). \tag{8.14}$$

Direct Beta integration gives

$$K_{V_{15}}(s) = \frac{415134720(s-1)}{(s+2)(s+4)(s+6)(s+8)(s+9)(s+10)(s+11)(s+12)(s+13)(s+14)(s+15)}, \tag{8.15}$$

where the denominator in (8.15) is the product of the displayed linear factors. Thus the Mellin detector is zero-free in the critical strip, and $v_1 = v_3 = v_5 = v_7 = 0$, together with $\int_0^1 V_{15} = 0$.

For $y = N + r$, $N = \lfloor y \rfloor \geq 1$, $0 \leq r < 1$, exact shifted Faulhaber summation gives

$$P_{V_{15}}(N + r) = \frac{N(N+1)}{(N+r)^{15}} Q_N(r). \tag{8.16}$$

The degree-thirteen Bernstein coefficients of Q_N, expressed in the falling-factorial basis $1, m, m^{\underline{2}}, m^{\underline{3}}, m^{\underline{4}}$ with $m = N - 1$, are all nonnegative rational numbers; Supplementary Table S2 records the complete array. Hence $P_{V_{15}}(y) \geq 0$ for every real $y \geq 1$, not merely at integer arguments. The resulting exact reciprocal-product form satisfies

$$T_{V_{15}}(x) = \frac{416S_9(\sqrt{x})^2}{x^9} - \frac{243232S_{11}(\sqrt{x})^2}{105x^{11}} + \frac{2912S_{13}(\sqrt{x})^2}{x^{13}} - \frac{115744S_{15}(\sqrt{x})^2}{255x^{15}} + T^{(0)}_{V_{15}}(x), \tag{8.17}$$

where the periodic coefficients in $T^{(0)}_{V_{15}}$ have zero mean. By Proposition 8.1 with $\ell = 4$, degree fifteen is minimal in this endpoint-flat polynomial class. The exact shifted Faulhaber convention and the corresponding minimality matrices are independently checked by the supplementary verifier.

The same construction continues one level further. The degree-eighteen detector

$$V_{18}(u) = \frac{u^2(1-u)^{11}}{8151}\big(8151u^5 + 3645u^4 + 198u^3 - 262u^2 - 77u - 7\big) \tag{8.18}$$

has $v_1 = v_3 = v_5 = v_7 = v_9 = 0$, zero mean, and a zero-free Mellin transform with denominator factors $s+2, s+4, s+6, s+8, s+10, s+11, \dots, s+18$. Its complete all-half-line Bernstein certificate is given in Supplementary Table S3. Thus the positive detector hierarchy reaches five removed odd residues, while the same signed quadratic-form obstruction remains.

The five-residue level is also minimal: Proposition 8.1 with $\ell = 5$ gives $\deg V \geq 18$. The corresponding rank computations and the agreement of all five Mellin transforms with (8.0b) are checked in the supplementary verifier.

Positivity of the canonical detector at every residue level has since been proved in [9] (Theorem 1 there), which credits [8] for necessity and the minimum-degree characterization. That result is not used here: the main theorem and the five finite certificates are independent of it.

9. Scope after prior-art comparison

The closest precedents separate into three classes. Classical Mertens, Riesz–Hardy–Littlewood, and general Möbius-convolution criteria already supply the analytic paradigm $K(s)/\zeta(s)$ and growth conditions equivalent to RH. Farey criteria supply genuinely nonnegative quadratic quantities and exact double arithmetic expressions, but their variables are Farey discrepancies or $M(x/m)$ with a gcd kernel, rather than a finite Möbius vector with an $x/(ab)$ kernel. The 2026 discrete-Laplace and local-Fourier criteria use different kernels and do not imply (3.2), (4.4), or the Section 8 finite certificates.

The contribution that survives this comparison is therefore narrow: the exact positive reciprocal-product kernel, its exact bounded-error transfer to a zero-free compact Mellin detector, the stated inertia results, and the five finite residue-removing detectors, each of minimal degree $3\ell + 3$ for $\ell \leq 5$ (Proposition 8.1, via [8, Theorem 2.1]). No general "universal detector" novelty is claimed, and no infinite detector hierarchy is proved.

10. Discussion

The main theorem replaces a one-dimensional Möbius detector by an exact finite quadratic form with entrywise nonnegative kernel. The growing inertia shows why this is not a positivity proof of RH. The exact signature in Section 6 concerns the primorial star; by interlacing (Corollary 6.1) it also yields unbounded negative inertia on the full squarefree support, where the Möbius coefficients are nonzero. The prime-square indices of Section 5 carry zero Möbius coefficients. Neither statement determines the sign or size of the form at the Möbius vector. Section 8 shows that positivity allows the first five RH-strength collective residues to be removed by explicit detectors, each of minimal degree $3\ell + 3$ (Proposition 8.1, via [8, Theorem 2.1]). Higher odd collective residues and the centered remainder remain. The unresolved problem is therefore to control this highly indefinite form on the particular Möbius vector while preserving cancellation between its collective and centered parts.

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Supplementary material

The supplement contains, in a separate tables file, the exact rational certificate tables for degrees twelve, fifteen, and eighteen. The script verify_detector_certificates.py reconstructs all five detectors from their definitions, verifies their Mellin transforms and moment cancellations, recomputes the shifted Faulhaber–Bernstein certificates, and checks the five minimality matrix ranks. Run

```
python3 verify_detector_certificates.py
```

The verifier requires SymPy. In an environment with `uv`, the isolated equivalent command is

```
uv run --with sympy python verify_detector_certificates.py
```

The supplement also contains `verify_reciprocal_kernel.py`, a Python standard library check of the exact truncation, cubic Faulhaber identity and transfer at nine fixed rational arguments, the indefinite two-by-two block, and the squarefree-star inertias for six and seven primes. These are finite reproduction checks; the general assertions rest on the analytic proofs above. Run `python3 verify_reciprocal_kernel.py` from the supplement directory.

Finally, `verify_degree_bound.py` (SymPy) checks that the five detectors have Mellin transforms of the form (8.0b) and checks the index condition of Corollary 6.1 for $6 \le m \le 30$.